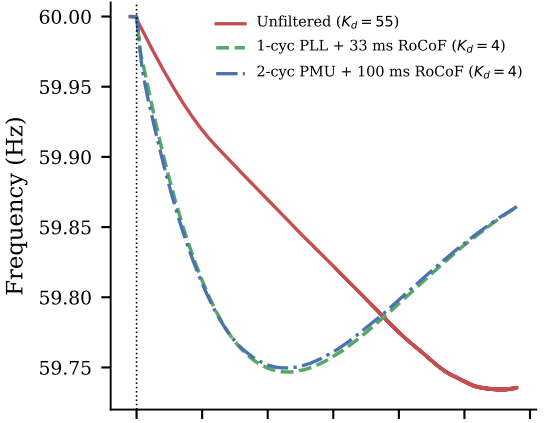
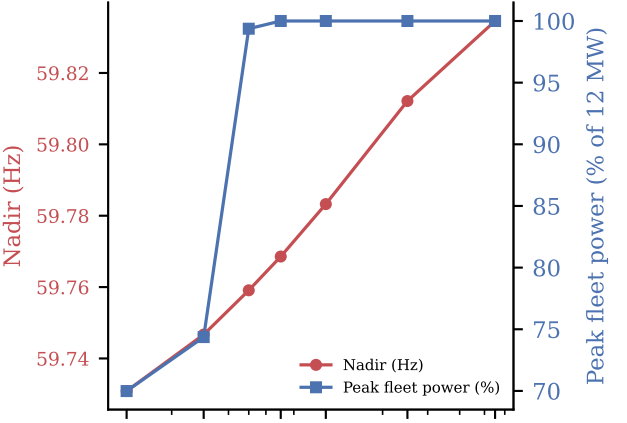

(a) PMU/PLL filtering & noise impact
Unfiltered ($K_d = 55$)
1-cyc PLL + 33 ms RoCoF ($K_d = 4$)
2-cyc PMU + 100 ms RoCoF ($K_d = 4$)
Frequency (Hz)
60.00
59.95
59.90
59.85
59.80
59.75
0
1000
2000
3000
4000
5000
6000
Time since contingency (ms)


(b) Gain vs. saturation regime
Nadir (Hz)
59.82
59.80
59.78
59.76
59.74
Peak fleet power (% of 12 MW)
100
95
90
85
80
75
70
Nadir (Hz)
Peak fleet power (%)
2
4
8
12
25
55
Derivative gain K_d (MW · s)